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Emergent Response Planning in LLMs

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Abstract

In this work, we argue that large language models (LLMs), though trained to predict only the next token, exhibit emergent planning behaviors: **their hidden representations encode future outputs beyond the next token**. Through simple probing, we demonstrate that LLM prompt representations encode global attributes of their entire responses, including *structure attributes* (e.g., response length, reasoning steps), *content attributes* (e.g., character choices in storywriting, multiple-choice answers at the end of response), and *behavior attributes* (e.g., answer confidence, factual consistency). In addition to identifying response planning, we explore how it scales with model size across tasks and how it evolves during generation. The findings that LLMs plan ahead for the future in their hidden representations suggest potential applications for improving transparency and generation control.

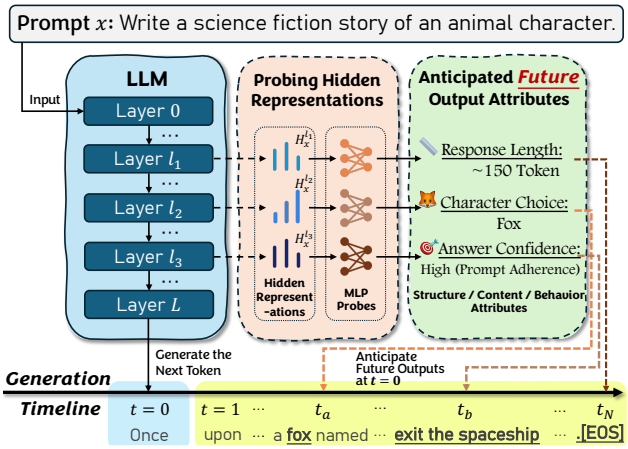


Figure 1. Illustration of probing LLMs for emergent response planning. After processing a prompt, hidden representations ($\mathbf{H}$) are extracted from the LLM's layers. MLP probes analyze these representations to predict future global attributes, including structure (length), content ("Fox"), and behavior attributes (answer confidence). LLMs can anticipate their future outputs—like the "Fox" appearing at t_a or the final length at t_N—long before generation.

1. Introduction

Large Language Models (LLMs) have demonstrated powerful capabilities across various tasks (Brown et al., 2020; Achiam et al., 2023; Touvron et al., 2023a; Anthropic, 2024). However, their next-token-prediction training objective leads to the view that they generate text through local, per-token prediction, without considering future outputs beyond the next immediate token (Bachmann & Nagarajan, 2024; Cornille et al., 2024) . This makes controlling the generation process challenging: *we are blind to the model's output tendency until keywords or the full response appear*. While prompt engineering and inference-time interventions (Liu et al., 2023; Li et al., 2024; Zhou et al., 2024) can guide responses, they lack insight and transparency into the model's internal plan for outputs.

In this work, we argue that LLMs, though trained to predict only the next token, display emergent planning behaviors: their hidden representations encode their future outputs beyond just the next token. Specifically, we observe that LLM prompt representations encode interesting global attributes of their upcoming responses. We call this phenomenon **response planning** and classify these global attributes into three categories: *structure attributes* (e.g., response length, reasoning steps), *content attributes* (e.g., character choices in storywriting, multiple-choice answers at the end of response), and *behavior attributes* (e.g., answer confidence, factual consistency).

We empirically identify response planning by training simple probes on LLM prompt representations to predict the global attributes of their upcoming responses. We find that these probes achieve non-trivial prediction accuracy, providing strong evidence that LLMs plan at least part of their entire response in advance when they read the prompt. Through further ablations, we find that planning abilities positively scale with model size, peak at the beginning and end of responses, share certain planning patterns across

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models, and exceed their self-verbalized awareness.

The contribution of our work is two-fold: (1) To our best knowledge, we introduce the first formal definition and framework of emergent response planning in LLMs. (2) We demonstrate empirically that LLMs perform emergent response planning through systematic probing experiments across various attributes types and tasks, and investigate their properties. These findings shed light on LLMs' internal mechanisms and suggest novel approaches for predicting and controlling outputs pre-generation, potentially enhancing model controllability.

2. Related Work

Understanding LLM hidden representations. LLM hidden representations encode more information than they actively use (Saunders et al., 2022; Burns et al., 2022). Patterns in these representations can be identified using linear or MLP probes (nostalgebraist, 2020; Li et al., 2022; Belrose et al., 2023; Zou et al., 2023; Ji et al., 2024) and leveraged to influence model behaviors such as truthfulness (Hernandez et al., 2023; Li et al., 2024), instruction-following (Heo et al., 2024), and sentiment (Turner et al., 2024). They are also useful for training additional regression or classification heads on transformer layers for tasks like reasoning (Han et al., 2024; Damani et al., 2024), high-dimensional regression (Tang et al., 2024), and harmful content detection (Rateike et al., 2023; MacDiarmid et al., 2024; Qian et al., 2024).

Our work also utilizes LLM hidden representations but differs in focus. Rather than using hidden states as feature extractors for external tasks, we probe model-generated data to understand how these states encode the model's own planning attributes during generation.

Prior works exploring response planning in LLM. Previous studies have examined whether LLMs can anticipate beyond the next token. Future Lens (Pal et al., 2023) models token distributions beyond the immediate next token using linear approximation. (Geva et al., 2023) studies how LLMs retrieve factual associations during generation, while (Men et al., 2024) extends this to Blocksworld planning, suggesting LLMs consider multiple planning steps simultaneously. (Pochinkov et al., 2024) finds that tokens at context-shifting positions may encode information about the next paragraph. (Wu et al., 2024) hypothesizes LLMs' lookahead capability and tests two mechanisms—pre-caching and breadcrumbs—in a myopic training setting.

While prior works examine relatively narrow aspects like predictions several tokens ahead or knowledge retrieval in specialized scenarios, our work delves deeper to reveal the broader response planning landscape of LLMs. We provide the first formal definition of response planning in LLMs, investigate comprehensive planning attributes, and demonstrate planning capabilities across diverse real-world tasks.

3. Emergent Response Planning in LLMs

If LLMs plan ahead for their entire response in prompt representations, then some global attributes of their upcoming responses can be predicted from the prompt, without generating any tokens. In this section, we first describe how existing probing techniques can investigate the global responses encoded in LLM prompt representations (Section 3.1). We then outline the setup for training our probes, including the response attributes of interest and the data collection pipeline (Section 3.2). Finally, we discuss experimental details before presenting our results (Section 3.3).

3.1. Probing for Future Responses

We study an L-layer decoder LLM $\pi(\mathbf{y} \mid \mathbf{x})$ that generates a response $\mathbf{y} = (y_1, \ldots, y_n)$ given a prompt $\mathbf{x} = (x_1, \ldots, x_m)$ sampled from a prompt distribution $p(\mathbf{x})$. During generation, the model encodes the input $(\mathbf{x} \circ \mathbf{y}_{1:t})$ into layer-wise representations $\{\mathbf{H}^l_{\mathbf{x}\circ\mathbf{y}_{1:t}}\}^L_{l-1}$, with the next token greedily decoded from the projection of final-layer representations $y_{t+1} = \arg\max(f_{\text{out}}(\mathbf{H}^L_{\mathbf{x}\circ\mathbf{y}_{1:t}}))$.

We investigate whether the prompt representations $\mathbf{H}^l_{\mathbf{x}}$, which produce the first response token y_1, also capture some global attributes of their upcoming response $\mathbf{y}$ (e.g., response length).

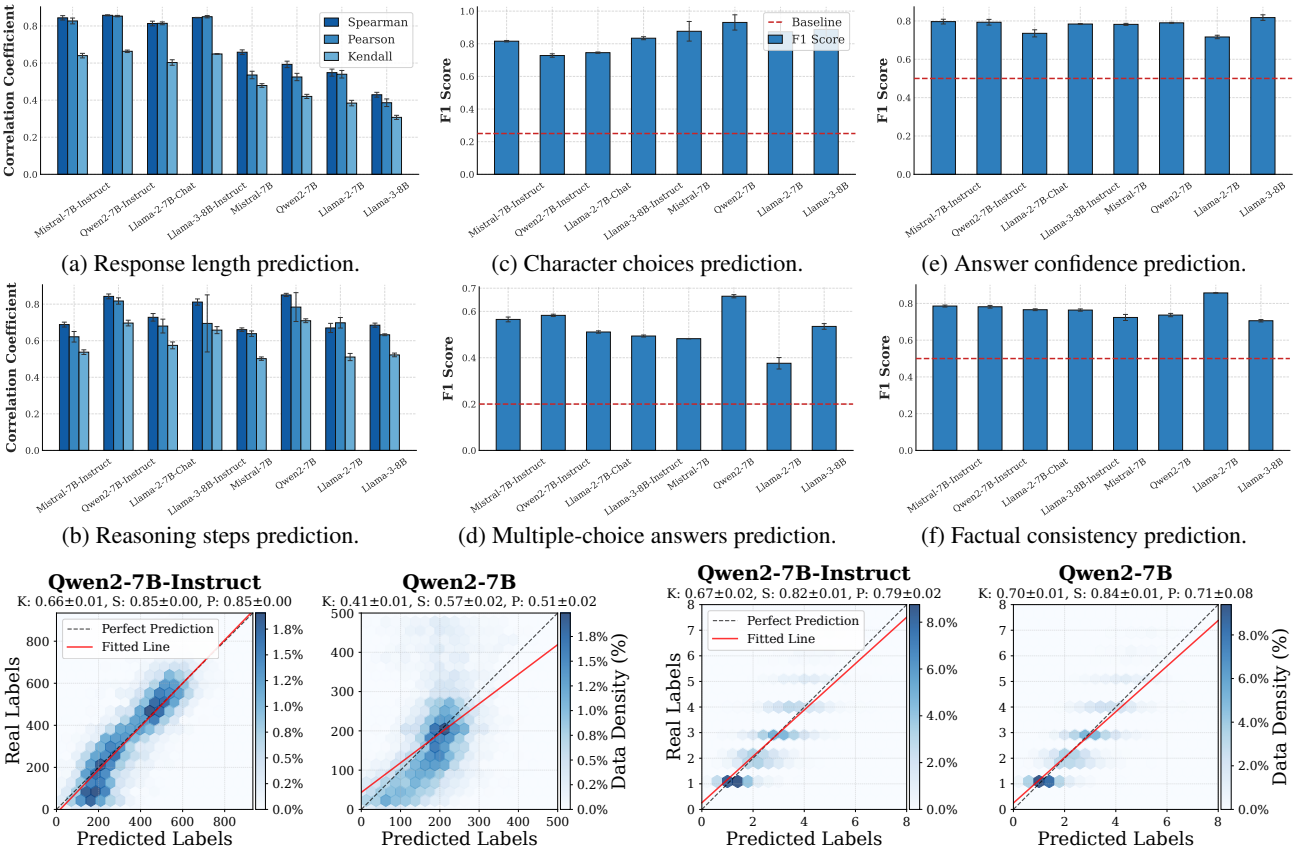
Formally, we define the *attribute rule* as $g(\mathbf{y})$, which summarizes the attributes from the generated responses (e.g., counting tokens in $\mathbf{y}$). Building on prior work on interpretability, if the prompt representations do capture these attributes, we can "probe" the hidden representations to predict the attributes without generating any response token: $h_\theta(\mathbf{H}^l_{\mathbf{x}}) \to g(\mathbf{y})$. If probing yields non-trivial predictions, we conclude that the LLM exhibits *response planning*.

3.2. Probing Setup

To study response planning in LLMs, we first design tasks $T = (p(\mathbf{x}), g(\mathbf{y}))$, consisting of a prompt distribution $p(\mathbf{x})$ eliciting key response attributes of interest $g(\mathbf{y})$ as probing targets. Next, we introduce the data collection pipeline for training probes.

Task design. The studied response attributes must be global, meaning they cannot be determined from the first response token and should ideally be distributed across the entire response. We focus on six tasks that elicit response attributes across three categories: structure, content, and behavior.

1. **Structure attributes** capture response-level features: the *response length prediction* prompts LLMs to fol-



(a) Response length prediction. (c) Character choices prediction. (e) Answer confidence prediction.

(b) Reasoning steps prediction. (d) Multiple-choice answers prediction. (f) Factual consistency prediction.

(g) Example fitting results for response length prediction. (h) Example fitting results for reasoning steps prediction.

Figure 2. Prediction results within the dataset. Regression tasks (response length, reasoning steps) show high accuracy and strong correlation with targets, as measured by Kendall (K), Spearman (S), and Pearson (P) coefficients. Classification tasks (character choices, multiple-choice answers, confidence, factual consistency) perform significantly above random baseline according to F1 scores. These results suggest that the model demonstrates emergent planning capabilities for future response attributes.

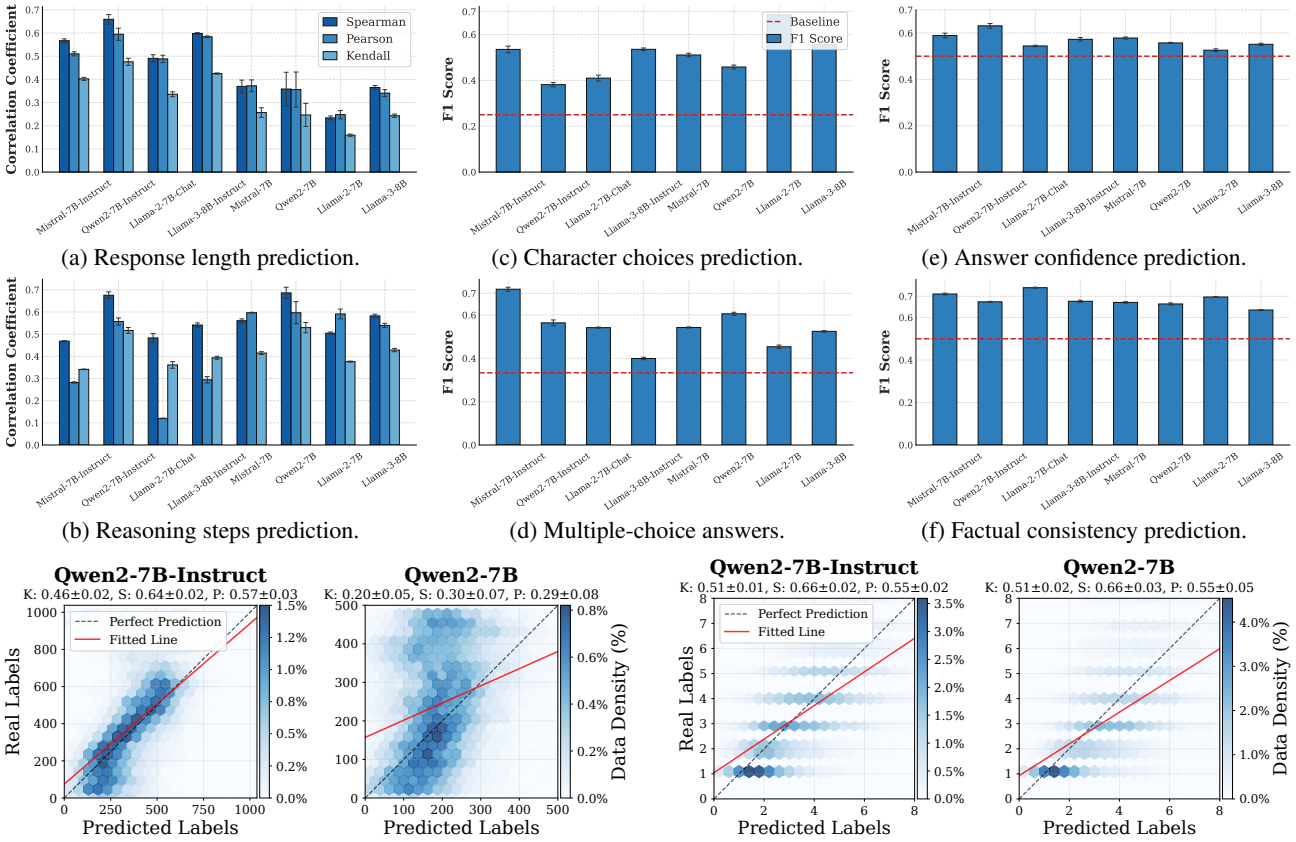
low human instructions, with the number of tokens counted as the probing target; the *reasoning steps prediction* prompts LLMs to solve math problems, with the number of reasoning steps as the probing target.

2. **Content attributes** track specific words appearing anywhere but not at the start of the response: *character choices prediction* prompts LLMs to write a story featuring an animal character, with the character choice as the probing target; *multiple-choice answers prediction* prompts LLMs to answer a question after reasoning (e.g., "please first explain then give your answer"), with the selected answer as the probing target.

3. **Behavior attributes** require external ground truth labels for validation: the *answer confidence prediction* prompts LLMs to answer challenging multiple-choice questions, with the correctness of answers judged by ground-truth labels as the probing target; the *factual consistency prediction* prompts LLMs to discuss and then agree/disagree with given statements, with the match between LLM's stance and statement ground-truth validity as the probing target.

Following the prompting strategies described in each task, we carefully pair datasets with corresponding prompts. We use prompts from Ultrachat (Ding et al., 2023) and AlpacaEval (Taori et al., 2023) for response length; GSM8K (Cobbe et al., 2021) and MATH (Saxton et al., 2019) for reasoning steps; TinyStories (Eldan & Li, 2023) and ROCStories (Mostafazadeh et al., 2016) for character choices; CommonsenseQA (Talmor et al., 2019) and SocialIQA (Sap et al., 2019) for multiple-choice answers; MedMCQA (Pal et al., 2022) and ARC-Challenge (Clark et al., 2018) for answer confidence; CREAK (Onoe et al., 2021) and FEVER (Thorne et al., 2018) for factual consistency. Please see Appendix A.3.1 for more details about task design.

Data collection. For each task $T = (p(\mathbf{x}), g(\mathbf{y}))$, we collect datasets for probing. We sample prompts $\mathbf{x}_i$ from the prompt distribution $p(\mathbf{x})$, store prompt representations $\mathcal{H}_i = \{\mathbf{H}^l_{\mathbf{x}_i}\}_{l=1}^{L}$, generate responses to the prompts $\mathbf{y}_i = \arg\max \pi(\mathbf{y} \mid \mathbf{x_i})$, and store probing targets $\hat{g}_i = g(\mathbf{y}_i)$.



(a) Response length prediction. (c) Character choices prediction. (e) Answer confidence prediction.

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(g) Example fitting results for response length prediction. (h) Example fitting results for reasoning steps prediction.

Figure 3. Cross-dataset generalization results. For regression tasks (response length, reasoning steps), correlations with targets remain strong despite reduced accuracy compared to in-dataset testing, as shown by Kendall (K), Spearman (S), and Pearson (P) coefficients. Classification tasks (character choices, multiple-choice, confidence, factual consistency) maintain above-baseline F1 scores. These results suggest the probes detect generalizable patterns rather than dataset-specific features, indicating transferable emergent planning capabilities within the task domain.

This creates a dataset of prompt representations and their future response attributes: $\mathcal{D} = \{\mathcal{H}_i, \hat{g}_i\}_{i=1}^{N}$. With this dataset, we then train a probe to predict targets from representations.

See Appendix A.3 for details on data collection, including task-specific and model-specific prompt templates, as well as data filtering and augmentation methods.

3.3. Experimental Details

Probe training. We train one-hidden-layer MLPs with ReLU activation, with hidden sizes chosen among $\mathcal{W} = \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024\}$. The output size is 1 for regression and the number of classes with a softmax layer for classification. Each probe is trained for 400 epochs using MSELoss for regression and CrossEntropyLoss for classification. Datasets are split $60 : 20 : 20$ for train-validation-test. We perform a grid search over MLP hidden sizes $\mathcal{W}$ and representation layers $\mathcal{H}$ (as inputs to the probes), reporting the test scores for the best hyperparameters. Results are averaged across three random seeds.

Probe evaluation. For regression tasks (response length and reasoning steps), we evaluate using Spearman, Kendall and Pearson correlation coefficients, which measure the strength and direction of monotonic (Spearman, Kendall) and linear (Pearson) relationships between predicted and target values. For classification tasks, we evaluate using F1 scores: 4-class classification for character choices, 5-class classification for multiple-choice answers, and binary classification for answer confidence and factual consistency. In our setup, accuracy aligns with F1 score for classification due to strict class balance across tasks.

Language models. We experiment with both instruction-tuned models (Llama-2-7B-Chat, Llama-3-8B-Instruct, Mistral-7B-Instruct, and Qwen2-7B-Instruct) and their corresponding base models (Llama-2-7B, Llama-3-8B, Mistral-7B, and Qwen2-7B). See Appendix A.1 for model details.